

ApexPlanet Cybersecurity Internship - Task 2

Vulnerability Assessment and Network Security Report

Objective

Perform reconnaissance, network scanning, firewall configuration and vulnerability assessment using professional cybersecurity tools.

Tools Used

WHOIS, NSLOOKUP, Google Dorking, Netcat, Nmap, iptables, Nessus Essentials.

Activities Performed

- WHOIS lookup on google.com
- DNS lookup using nslookup
- Google Dorking queries
- Banner grabbing using Netcat
- Nmap scans (Basic, Version Detection, SYN, UDP and OS Detection)
- Firewall configuration using iptables
- Installation and execution of Nessus Essentials
- Localhost vulnerability assessment

Screenshots

Screenshot 2026-06-26 123027.png

```
dell@DESKTOP-0477P85:~$ cd /ApexplanetLab
-bash: /ApexplanetLab: No such file or directory
dell@DESKTOP-0477P85:~$ cd /ApexplanetLab
-bash: cd: /ApexplanetLab: No such file or directory
dell@DESKTOP-0477P85:~$ cd
```

Screenshot 2026-06-26 123430.png

```
dell@DESKTOP-0477P85:~/ApexplanetLab$ whois google.com
Domain Name: GOOGLE.COM
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.co
m
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#c
lientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp
#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#c
lientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#s
erverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp
#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#s
erverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
DNSSEC: unsigned
```

Screenshot 2026-06-26 123445.png

to be accurate,
the data is provided "as is" with no guarantee or warranties regarding its accuracy.

By submitting a WHOIS query, you agree that you will use this data only for lawful purposes and that, under no circumstances will you use this data to:

(1) allow, enable, or otherwise support the transmission by email, telephone, or facsimile of mass, unsolicited, commercial advertising, or spam; or

(2) enable high volume, automated, or electronic processes that send queries, data, or email to MarkMonitor (or its systems) or the domain name contacts (or its systems).

MarkMonitor reserves the right to modify these terms at any time.

By submitting this query, you agree to abide by this policy.

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Contact us at +1.8007459229

In Europe, at +44.02032062220

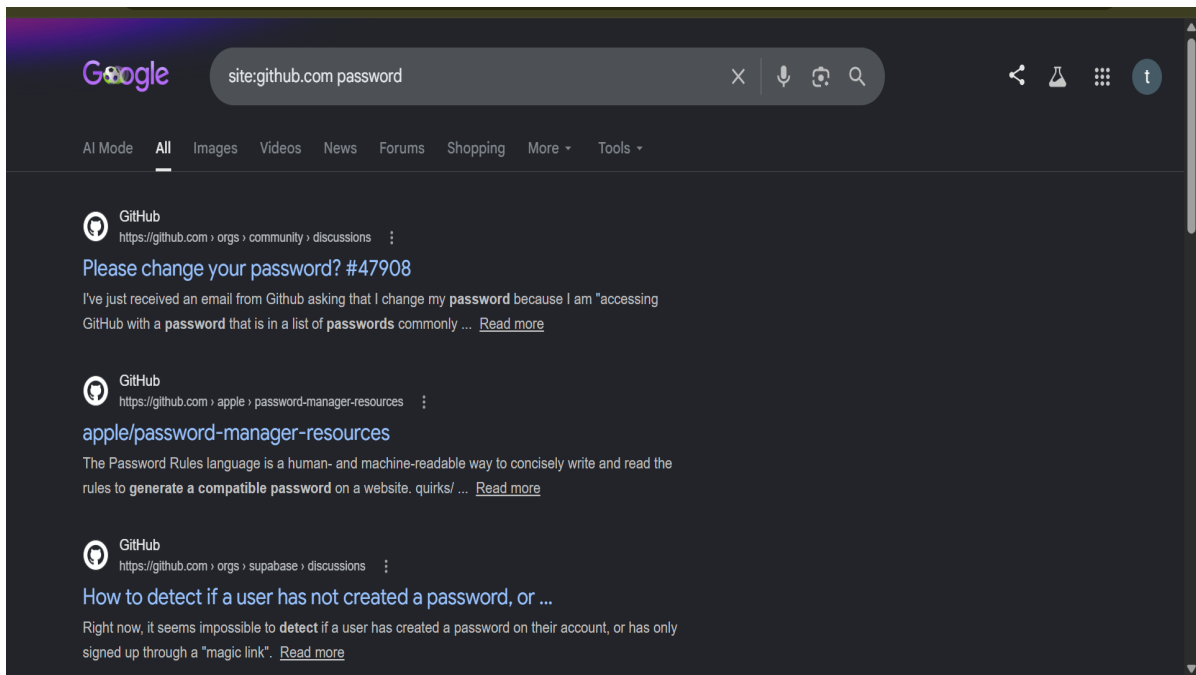
Screenshot 2026-06-26 123542.png

```
dell@DESKTOP-0477P85:~/ApexplanetLab$ nslookup google.com
Server:      10.255.255.254
Address:     10.255.255.254#53
```

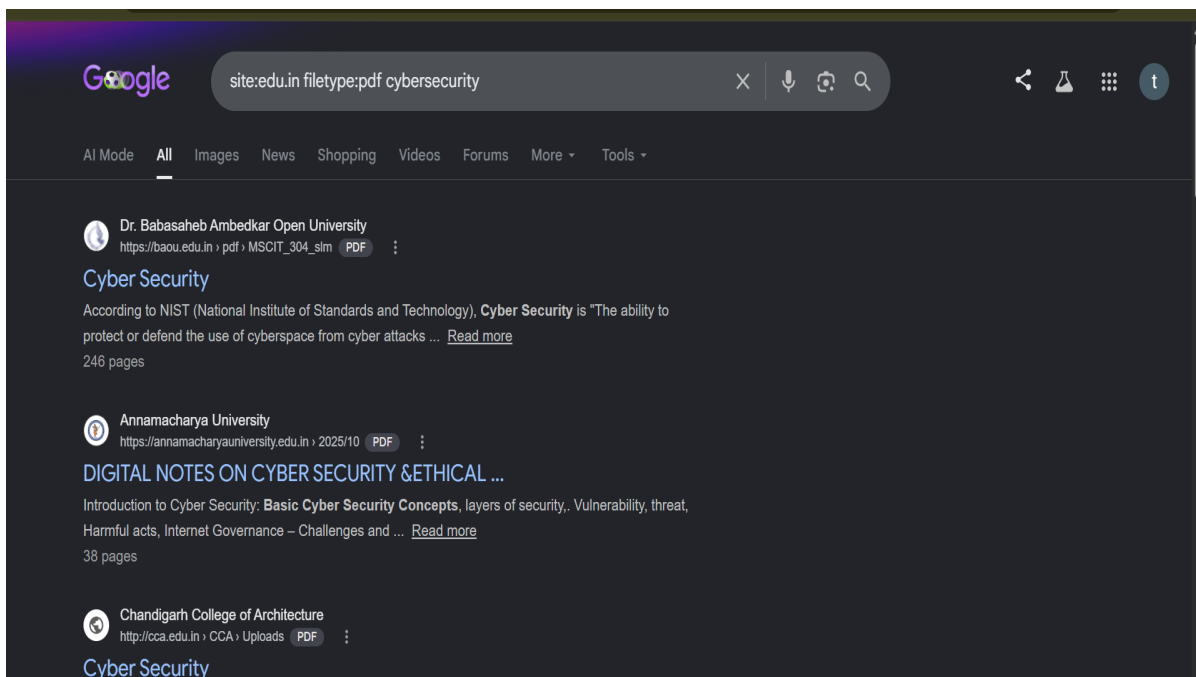
Non-authoritative answer:

```
Name:   google.com
Address: 192.178.173.102
Name:   google.com
Address: 192.178.173.139
Name:   google.com
Address: 192.178.173.101
Name:   google.com
Address: 192.178.173.113
Name:   google.com
Address: 192.178.173.100
Name:   google.com
Address: 192.178.173.138
Name:   google.com
Address: 2404:6800:4000:101d::8b
Name:   google.com
Address: 2404:6800:4000:101d::65
Name:   google.com
Address: 2404:6800:4000:101d::64
Name:   google.com
Address: 2404:6800:4000:101d::8a
```

Screenshot 2026-06-26 123722.png



Screenshot 2026-06-26 123803.png



Screenshot 2026-06-26 124023.png

```
dell@DESKTOP-0477P85:~/ApexplanetLab$ nc google.com 80
HEAD / HTTP/1.1
Host:google.com

HTTP/1.1 301 Moved Permanently
Location: http://www.google.com/
Content-Type: text/html; charset=UTF-8
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-DsnozafZSdU8RSFkJqiQmg' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http:;report-uri https://csp.withgoogle.com/csp/gws/other-hp
Date: Fri, 26 Jun 2026 07:10:08 GMT
Expires: Sun, 26 Jul 2026 07:10:08 GMT
Cache-Control: public, max-age=2592000
Server: gws
Content-Length: 219
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN
```

Screenshot 2026-06-26 124250.png

```
dell@DESKTOP-0477P85:~/ApexplanetLab$ nmap localhost
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-26 07:12 +0000
0
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000081s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (conn-refused)

Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds
```

Screenshot 2026-06-26 124338.png

```

dell@DESKTOP-0477P85:~/ApexplanetLab$ nmap -sv localhost
Nmap 7.98 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255
  .1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discover
  y to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery
  probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: someti
  mes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags

```

Screenshot 2026-06-26 124423.png

```

dell@DESKTOP-0477P85:~/ApexplanetLab$ sudo nmap -o localhost
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-26 07:14 +0000
0
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.04 seconds
dell@DESKTOP-0477P85:~/ApexplanetLab$

```

Screenshot 2026-06-26 124457.png

```
dell@DESKTOP-0477P85:~/ApexplanetLab$ sudo nmap -sS localhost
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-26 07:14 +0000
0
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000011s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored s
tates.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.23 seconds
```

Screenshot 2026-06-26 124529.png

```
dell@DESKTOP-0477P85:~/ApexplanetLab$ sudo nmap -sU localhost
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-26 07:15 +0000
0
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000060s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored s
tates.
Not shown: 1000 closed udp ports (port-unreach)

Nmap done: 1 IP address (1 host up) scanned in 0.25 seconds
```

Screenshot 2026-06-26 124704.png

```
dell@DESKTOP-0477P85:~/ApexplanetLab$ nmap -sV localhost
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-26 07:16 +0000
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00011s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored s
tates.
Not shown: 1000 closed tcp ports (conn-refused)

Service detection performed. Please report any incorrect results
at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.42 seconds
```

Screenshot 2026-06-26 124843.png

```
Nmap done: 1 IP address (1 host up) scanned in 0.42 seconds
dell@DESKTOP-0477P85:~/ApexplanetLab$ sudo nmap -O localhost
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-26 07:18 +0000
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000075s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored s
tates.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS detail
s
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at h
https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.60 seconds
```

Screenshot 2026-06-26 124931.png


```
dell@DESKTOP-0477P85: ~/ApexPlanetLab$ sudo iptables -L
sudo: 'iptables': command not found
```

Screenshot 2026-06-26 125121.png

```
dell@DESKTOP-0477P85:~/ApexplanetLab$ which iptables
dell@DESKTOP-0477P85:~/ApexplanetLab$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [247 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [64.2 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [219 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [38.6 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:8 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 c-n-f Metadata [3996 B]
Fetched 848 kB in 2s (431 kB/s)
15 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Screenshot 2026-06-26 125225.png

```
Processing triggers for libc-bin (2.43-2ubuntu2) ...
dell@DESKTOP-0477P85: ~/ApexplanetLab$ iptables --version
iptables v1.8.11 (nf_tables)
dell@DESKTOP-0477P85: ~/ApexplanetLab$
```

Screenshot 2026-06-26 125313.png

```
dell@DESKTOP-0477P85:~/ApexplanetLab$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
dell@DESKTOP-0477P85:~/ApexplanetLab$
```

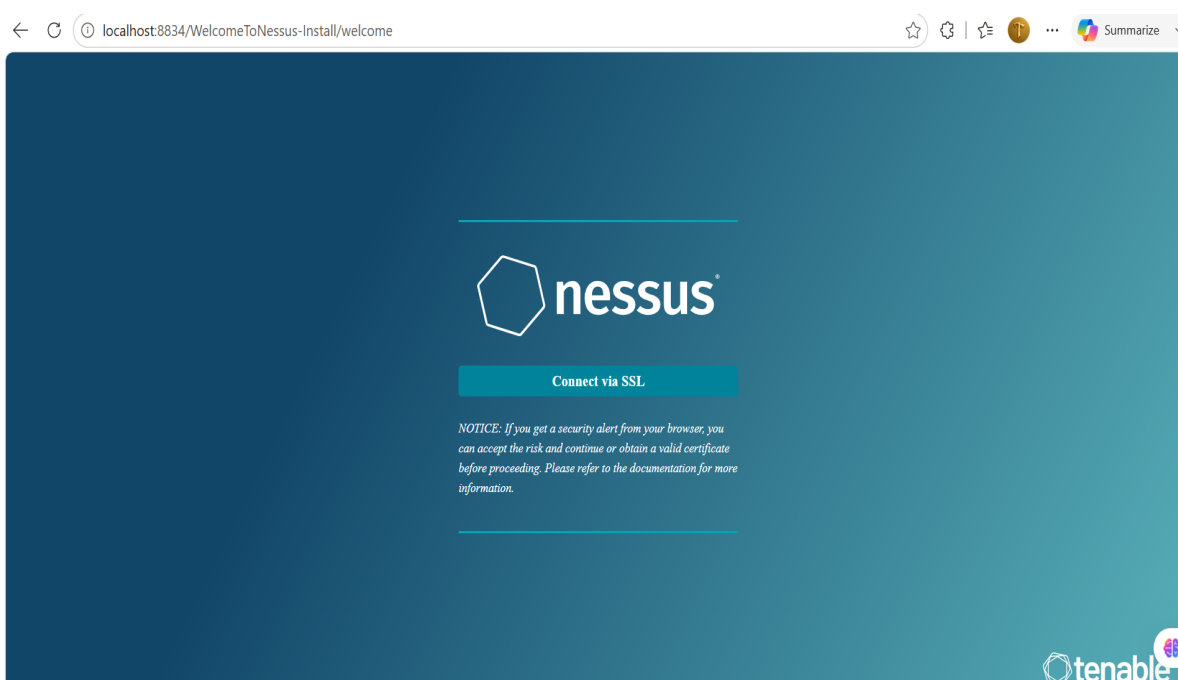
Screenshot 2026-06-26 125435.png

```
dell@DESKTOP-0477P85:~/ApexplanetLab$ sudo iptables -A INPUT -p
tcp --dport 80 -j DROP
dell@DESKTOP-0477P85:~/ApexplanetLab$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination
DROP      tcp  --  anywhere              anywhere             tcp
p dpt:http

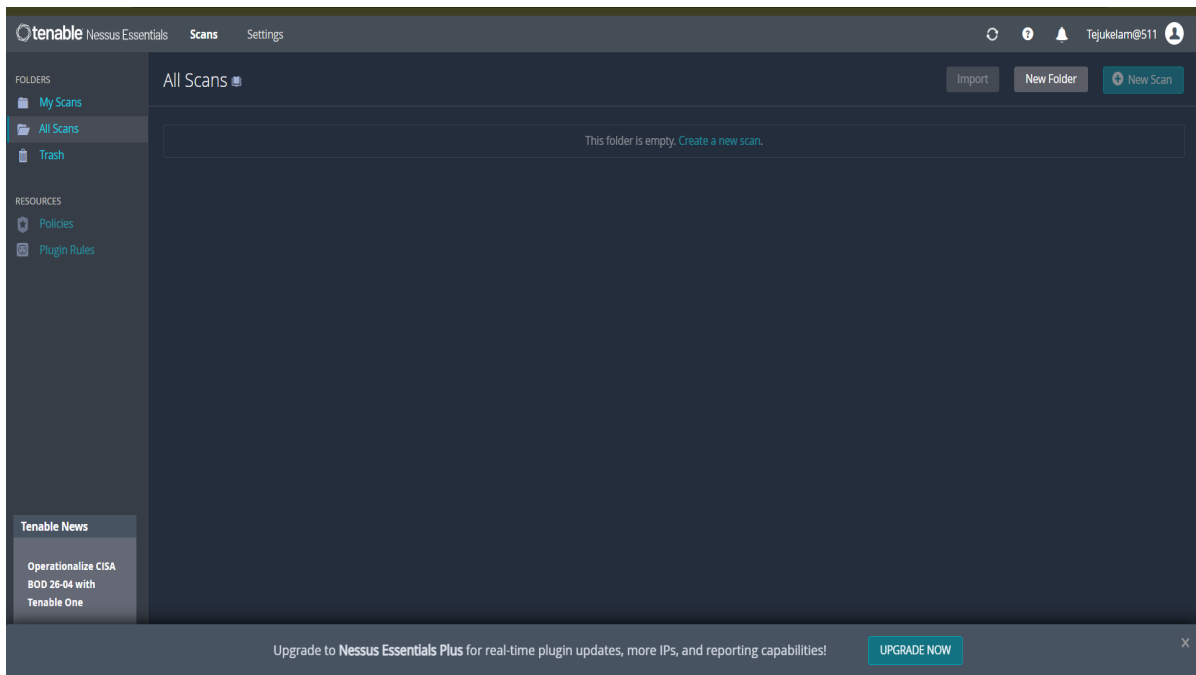
Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
```

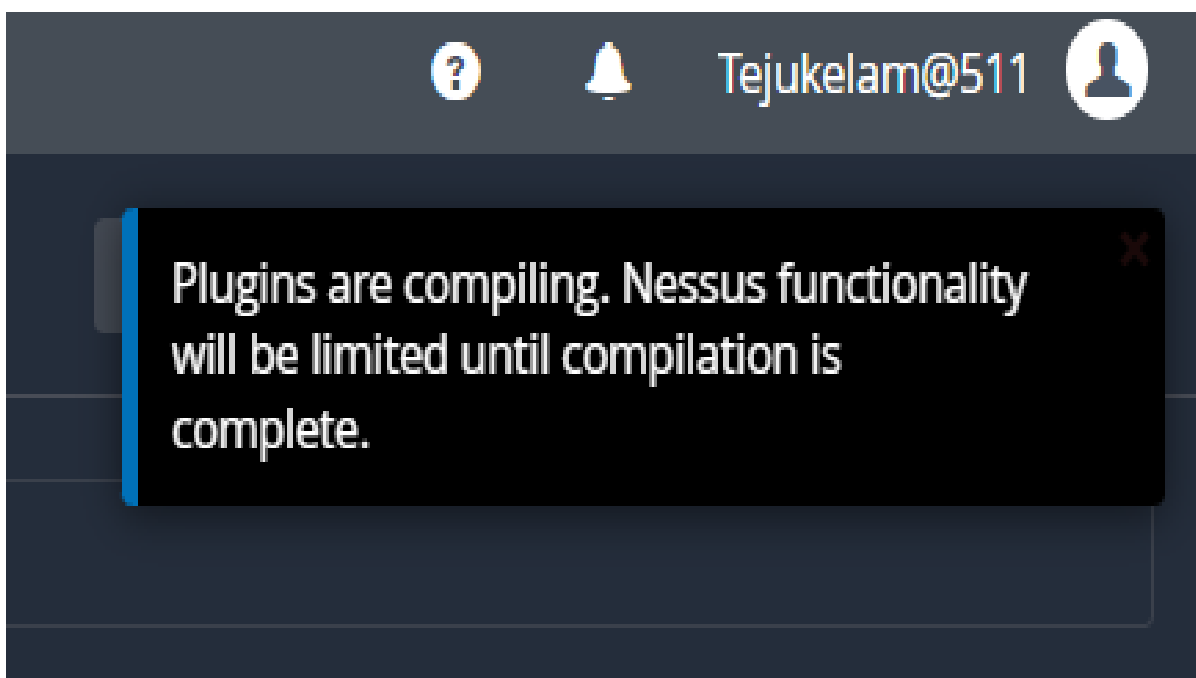
Screenshot 2026-06-26 145739.png



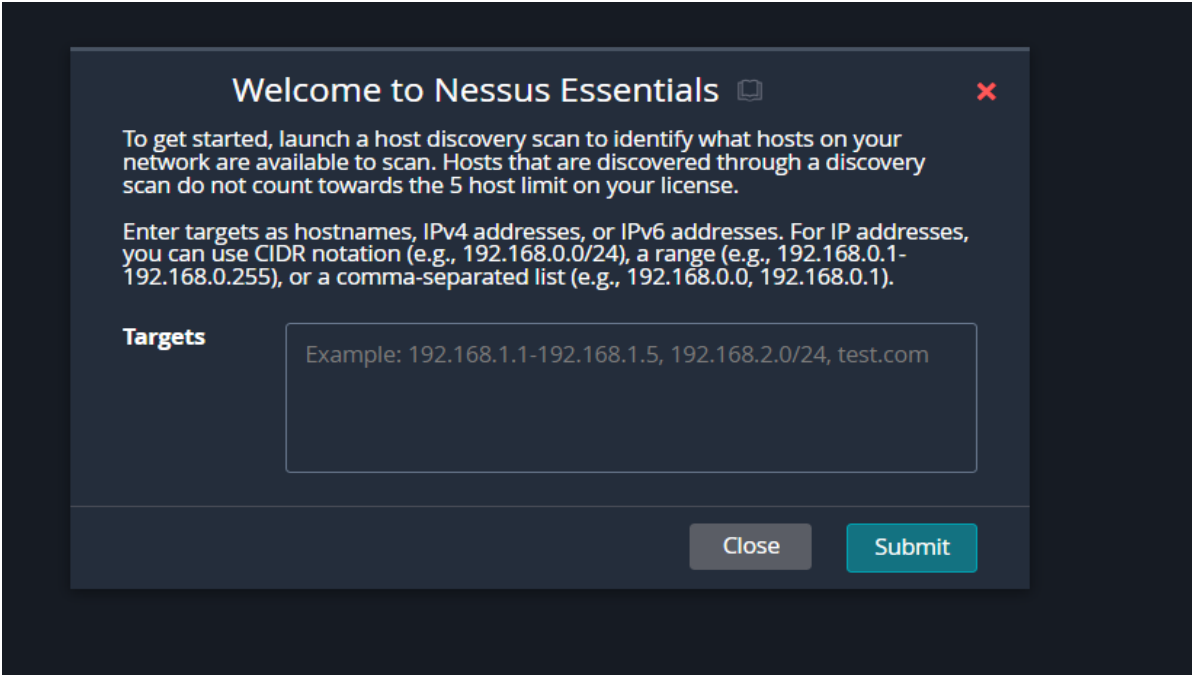
Screenshot 2026-06-26 160010.png



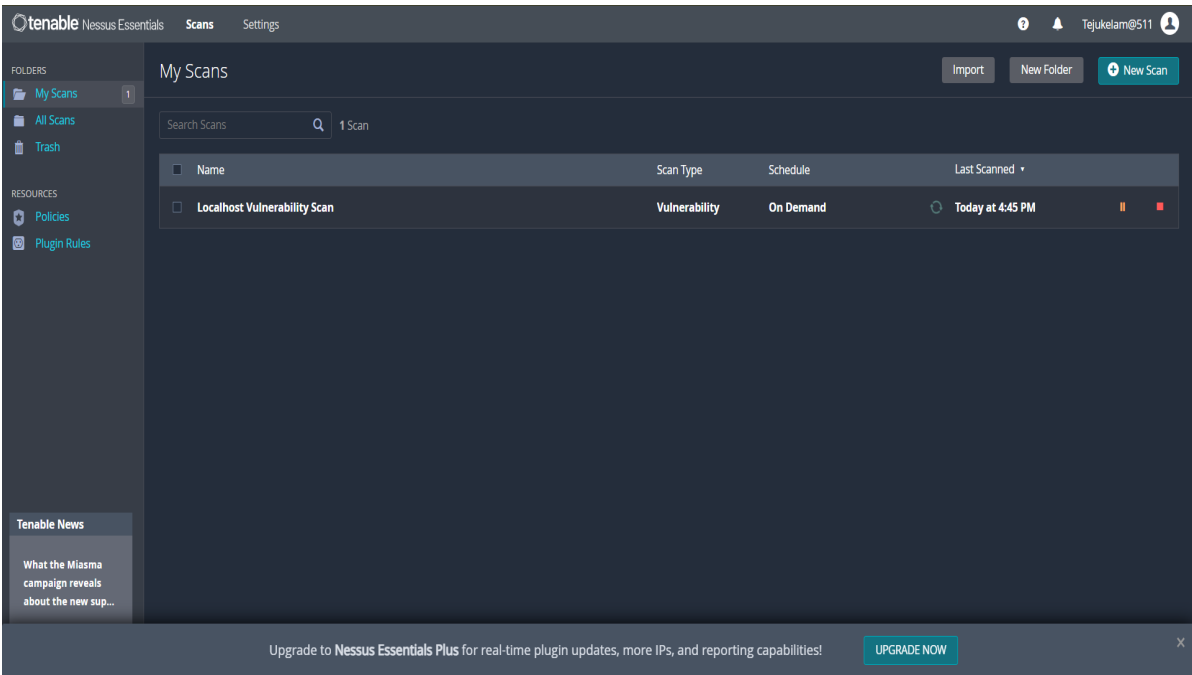
Screenshot 2026-06-26 160303.png



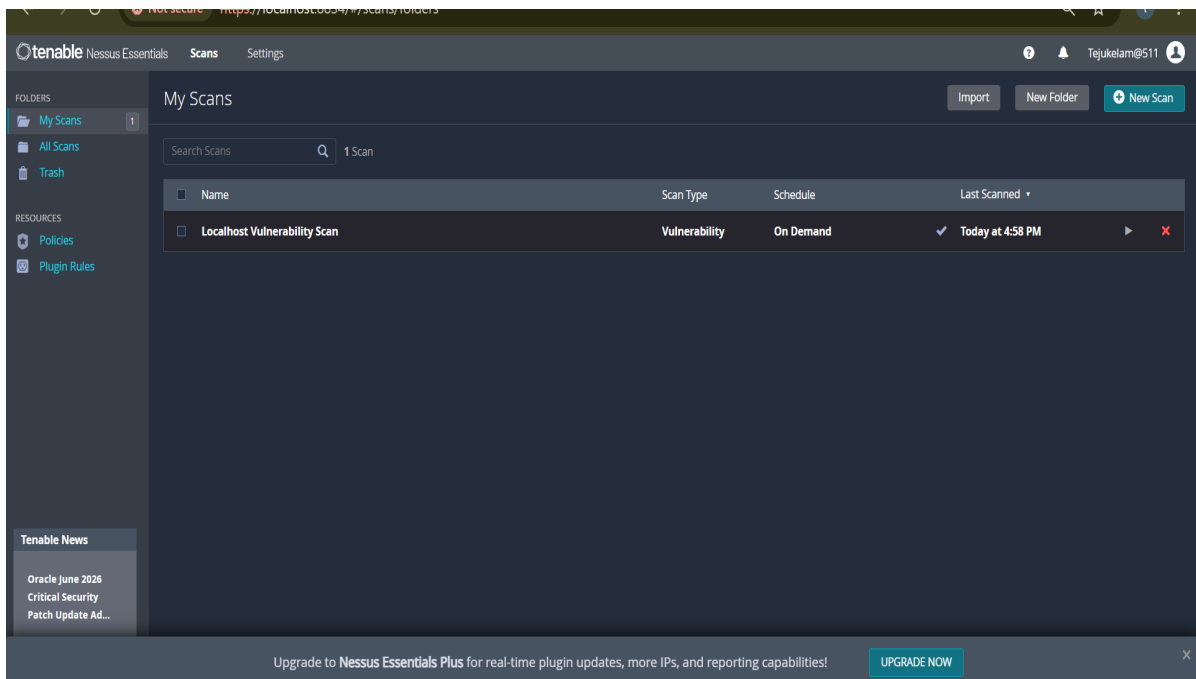
Screenshot 2026-06-26 163112.png



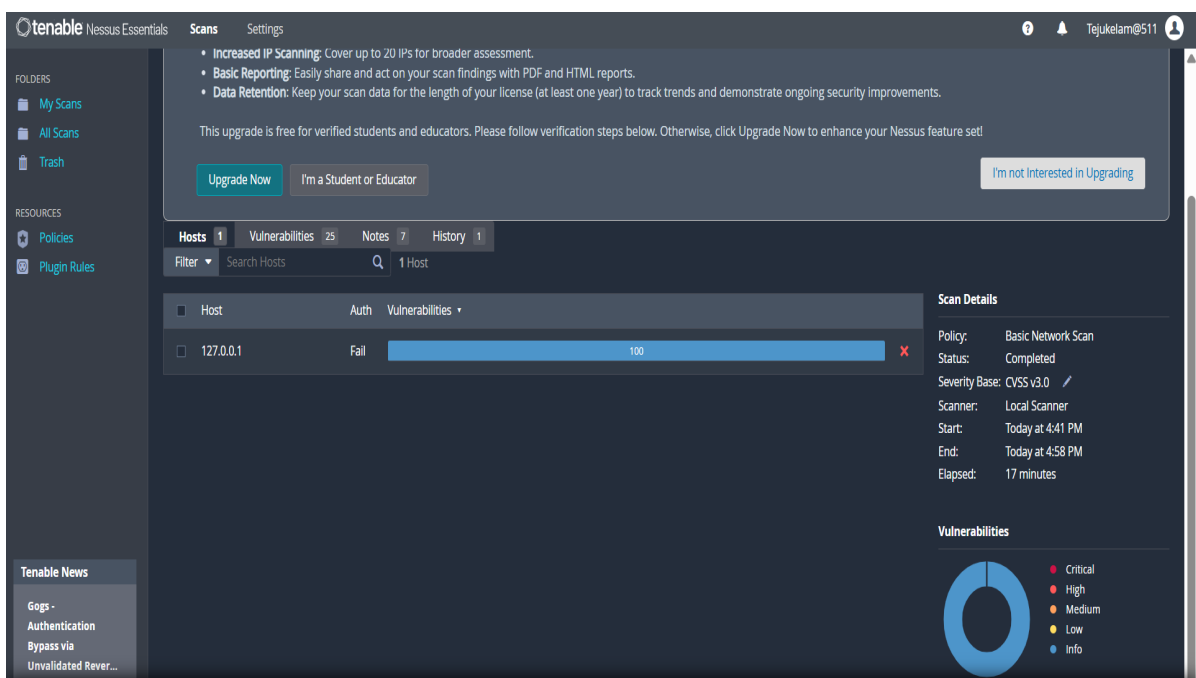
Screenshot 2026-06-26 164519.png



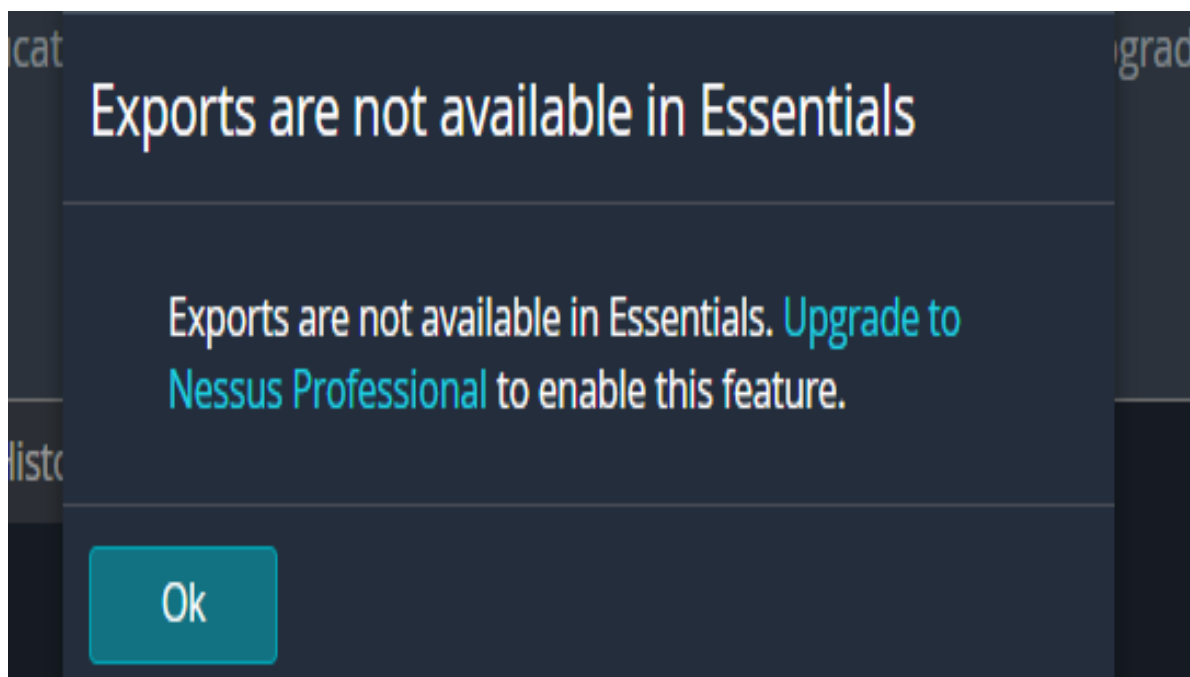
Screenshot 2026-06-26 170105.png



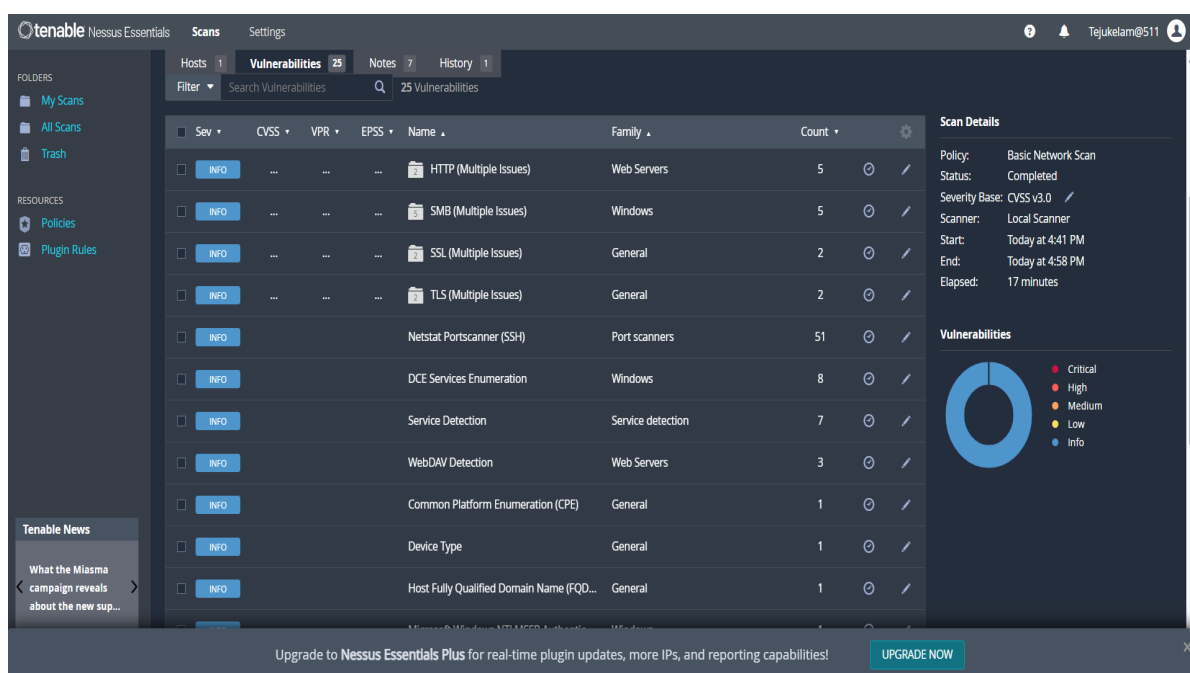
Screenshot 2026-06-26 170237.png



Screenshot 2026-06-26 170345.png



Screenshot 2026-06-26 170521.png



Conclusion

The assessment was successfully completed. Reconnaissance, network scanning, firewall rule configuration and Nessus vulnerability scanning were performed successfully. The scan identified informational findings on localhost and demonstrated practical usage of industry-standard cybersecurity tools.